

FULL STACK WEB DEVELOPMENT - SOURCE CODE

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Question 4: DOM Demonstration (HTML, CSS, JavaScript)

File: index.html

```
<!DOCTYPE html>
<!-- Program: DOM Demonstration
      Purpose: Show how JavaScript changes the content, style and attributes
              of HTML elements dynamically using the DOM. -->
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>DOM Demonstration</title>
  <!-- CSS file -->
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <!-- Elements that JavaScript will modify (each has an id) -->
  <div id="box">
    <h1 id="heading">Original Heading</h1>
    <p id="para">This is the original paragraph text.</p>
    <a id="link" href="https://www.example.com">Original link</a>
    <br><br>
    <input type="text" id="nameField" placeholder="Original placeholder">
  </div>

  <!-- Buttons that trigger the DOM changes -->
  <button id="btnContent">Change Content</button>
  <button id="btnStyle">Change Style</button>
  <button id="btnAttr">Change Attributes</button>
  <button id="btnReset">Reset</button>

  <div id="info">Click a button to modify the DOM.</div>

  <!-- JavaScript file -->
  <script src="script.js"></script>
</body>
</html>
```

File: style.css

```
/* style.css - DOM Demonstration */

body { font-family: Arial, sans-serif; margin: 30px; background: #f7f7f7; }
/* Container box whose style will be changed by JavaScript */
#box {
  background: #ffffff;
  border: 2px solid #999;
  padding: 20px;
  width: 520px;
  border-radius: 6px;
}
button { padding: 8px 12px; margin: 10px 6px 0 0; cursor: pointer; }
/* Area that shows a message after every action */
#info {
  margin-top: 16px;
  padding: 10px;
  width: 520px;
  background: #eef;
  font-family: monospace;
  font-size: 13px;
  box-sizing: border-box;
}
```

File: script.js

```
// script.js - DOM Demonstration

// Access the elements using the DOM (getElementById returns the element)
const box      = document.getElementById("box");
const heading  = document.getElementById("heading");
const para     = document.getElementById("para");
const link     = document.getElementById("link");
const field    = document.getElementById("nameField");
const info     = document.getElementById("info");

// 1. Change CONTENT
function changeContent() {
  // textContent: sets plain text only
  heading.textContent = "Heading Changed by JavaScript";
  // innerHTML: sets text that can contain HTML tags (<b>, <i>)
  para.innerHTML = "The paragraph now uses <b>innerHTML</b> " +
    "to add <i>formatted</i> text.";
  link.textContent = "Visit MDN Web Docs";
  info.textContent = "Content changed using textContent and innerHTML.";
}

// 2. Change STYLE
function changeStyle() {
  // element.style.property changes the inline CSS of the element
  box.style.backgroundColor = "#fff3cd";
  box.style.border = "3px dashed #d35400";
  heading.style.color = "#c0392b";
  heading.style.fontSize = "36px";
  para.style.fontStyle = "italic";
  info.textContent = "Style changed using the element.style property.";
}

// 3. Change ATTRIBUTES
function changeAttributes() {
  // setAttribute(name, value) adds or changes an attribute
  link.setAttribute("href", "https://developer.mozilla.org");
  link.setAttribute("title", "Mozilla Developer Network");
  link.setAttribute("target", "_blank");
  field.setAttribute("placeholder",
    "Placeholder changed by setAttribute()");
  field.setAttribute("maxlength", "20");
  // getAttribute(name) reads the new values, shown in the info box
  info.textContent = "href = " + link.getAttribute("href") +
    " | title = " + link.getAttribute("title") +
    " | maxlength = " + field.getAttribute("maxlength");
}

// 4. Reset everything to the original state
function resetAll() {
  heading.textContent = "Original Heading";
  para.textContent = "This is the original paragraph text.";
  link.textContent = "Original link";
  // removeAttribute("style") removes all inline styles we added
  box.removeAttribute("style");
  heading.removeAttribute("style");
  para.removeAttribute("style");
  link.setAttribute("href", "https://www.example.com");
  link.removeAttribute("title");
  link.removeAttribute("target");
  field.setAttribute("placeholder", "Original placeholder");
  field.removeAttribute("maxlength");
  info.textContent = "DOM reset to original state.";
}

// Event listeners: run a function when its button is clicked
const byId = (id) => document.getElementById(id);
byId("btnContent").addEventListener("click", changeContent);
byId("btnStyle").addEventListener("click", changeStyle);
byId("btnAttr").addEventListener("click", changeAttributes);
byId("btnReset").addEventListener("click", resetAll);
```